COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling for Data Science

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Lectures	Tue and Thu 5:40 – 6:55 pm 301 Uris
Textbook Materials	• Wasserman "All of Statistics" • ISLR • Boos D., Stefanski L.A., Essential Statistical Inference: Theory and Methods, • Lecture notes as pdf
Grading	Data Challenges = 15% Homework = 25% Midterm = 20% Final Exam = 40%

Course Outline

Estimation (AOS Chapters 6, 9)

- Convergence
- Sample mean and higher-order moments
- Likelihood
- Confidence Intervals

Hypothesis Testing (Chapter10)

- Null and alternative
- Significance level
- Rejection region
- Types of errors
- Likelihood ratio test
- Connection with confidence intervals

Multivariate Normal and Exponential Family (Chapter 14)

- Sufficient statistics
- Properties of multivariate normal distribution

Regression (ISLR Chapter 3, 6)

- Multiple linear model
- Least squares method
- Model selection and regularization

Nonlinear models (Chapters 5, 7)

- GLM
- Proportions
- Counts
- Inference and estimation
- Nonparametric regression

Bayesian models (Chapters 11, 24)

- Priors
- MCMC
- Bayesian computation

If time permits

- Missing data

Introduction and Review of Probability Theory

Course Goals:

- Review a very broad collection of methods so you are introduced to the most fundamental statistical procedures.
- Provide you with both theoretical and practical background, including real data examples.
- Understand the assumptions and limitations of statistical methods.
- Being able to select the appropriate method for analyzing a specific dataset.

What is Statistics?

- A branch of mathematics that learns from data
- *Pause*: Then what is machine learning??
- Applied statistics: concerned with methods for collecting, analyzing, and drawing conclusions from data.
- Theoretical statistics: establishes the properties and conditions under which the data analysis methods work.
- Without rudimentary knowledge of how it works, people can't make evaluations of a wide variety of numerical encounters in daily life.
- Many statistical methods are relevant and useful in a wide range of academic and non-academic disciplines, beyond just the statistical and data sciences.

Example

Data collected in 19th century by Darwin who studied the effect of pollination on Zea mays plants (aka maize or corn). See `01-Introduction.R` file

	pot	pair	type	height
1	I	a	cross	23.500
2	I	a	self	17.375
3	I	b	cross	12.000
4	I	b	self	20.375
5	I	c	cross	21.000
6	I	c	self	20.000
7	II	d	cross	22.000
8	II	d	self	20.000
9	II	e	cross	19.125
10	II	e	self	18.375

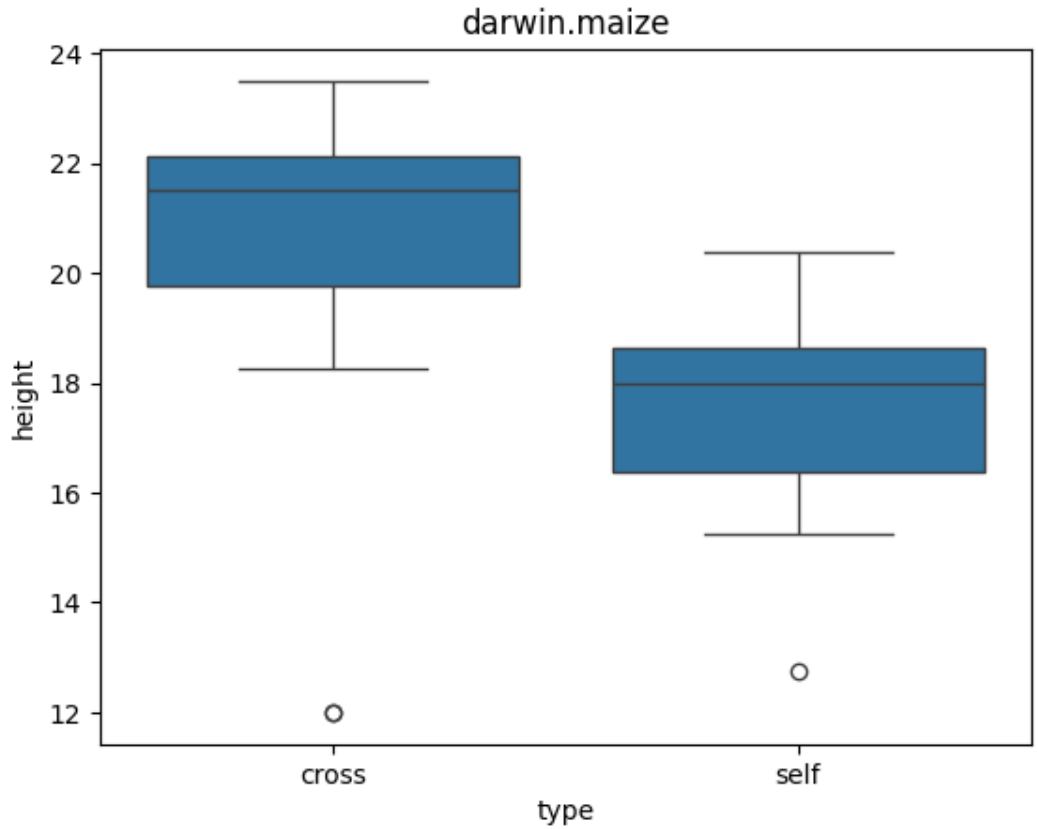
…

Research question:

Do cross-fertilized plants tend to grow taller?

How do we answer this question scientifically?

- You can examine the difference between the two groups visually.
- Or you can write a probabilistic model using random variables and then test a statistical hypothesis.



Perform a hypothesis test:

```
> t.test(d2$diff)

        One Sample t-test

data:  d2$diff
t = 2.148, df = 14, p-value = 0.0497
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.003899165 5.229434169
sample estimates:
mean of x
 2.616667
```

What do these results mean and how was this test developed?

Key idea in statistics:

explain the variation

- To do this we need to express scientifically the variation in the data as observations of random variables and use a statistical model to explain it (at least partially). Statistical analysis requires data, which may be viewed as an observed sample from a population.

- The data $X_1, \ldots, X_n$ are considered as n, usually independent and identically distributed, random variables with a common population distribution/model. This population can be specified as a formula with either its cumulative distribution function (cdf) $F(x)$ or its probability density function (pdf) $f(x)$.

Back to the maize example

- Let a typical height of the self-pollinated plant be denoted Y
- Let a typical height of the cross-pollinated plant be denoted X
- Then one possible model is:

$$Y = \mu + \sigma\varepsilon$$
$$X = \mu + \eta + \sigma\varepsilon$$

- $\mu, \eta,$ and σ have unknown values called *model parameters.* They need to be *learned* from the data.
- ε is a random variable, called *error* or *noise*
- The research question is now: can we demonstrate that $\eta > 0$?
- There are always assumptions made to answer the research question. For example, $\varepsilon \sim N(0,1)$.

Back to the maize example (continued):

- In the context of a sample from a population, X and Y will have indexes to indicate that.
- Also, it is very common to use indexes for the different groups (in this case two groups: self and cross-pollinated).
- Then a more rigorously written model would be:

$$Y_i = \mu_1 + \sigma\varepsilon_{1i}, \qquad i = 1, \dots, n_1$$
$$X_i = \mu_2 + \sigma\varepsilon_{2i}, \qquad i = 1, \dots, n_2$$

Research question would be: can we show that $\mu_2 > \mu_1$?

There are more details: Darwin controlled the effect of temperature and humidity by planting in the same pot several pairs of cross-pollinated and self-pollinated plants. This means we must consider such dependence (see R file).

Maize example continued:

- We can estimate population average μ with the sample average $\bar{Y}$

$$\bar{Y} = \frac{Y_1 + \cdots + Y_{15}}{15} = 20.19167$$

- We can estimate population average $\mu + \eta$ with sample average $\bar{X}$

$$\bar{X} = \frac{X_1 + \cdots + X_{15}}{15} = 17.575$$

- Then to answer the research question we intuitively estimate η with

$$20.19167 - 17.575 = 2.616667$$

So, the estimated typical difference between cross and self-pollinated plants is about 2.6 inches and is positive, but …

- How accurate/reliable is this estimate?
- Can we safely conclude it is positive not by random chance, but due to systematic difference between the two groups?

Some Answers

- To ”prove” there is a systematic difference, we need to formalize the model and then use probability theory results.
- The answer will clearly depend on the sample size n and is usually more reliable as n increases.
- The assumptions play a key role in the trustworthiness of the result. Are the sample observations independent and identically distributed (iid); is the population normally distributed; …

Review of Important Probability Theory Results

Random Sample

- The *observed* sample is a collection of numbers, also known as the observations, the data or the data set. These are represented using lowercase letters, for example, $x_1, \ldots, x_n$.
- The *random* sample is a collection of random variables whose properties form our model for the data.

Let $X_1, \ldots, X_n$ be iid random sample from a population with pdf $f(x)$.

This means:

1. Each X_i, $i = 1, \ldots, n$ is a random variable.
2. Each X_i, $i = 1, \ldots, n$ has the same pdf $f(x)$.
3. $X_1, \ldots, X_n$ are mutually independent.

Mean and Variance (Chapter 3)

Usually, we denote the expected value (average or mean) of each X_i with μ and its variance with σ^2. This means:

$$\mathrm{E}(X) = \int x f(x) dx = \mu$$

$$\mathrm{Var}(X) = \int (x-\mu)^2 f(x) dx = \sigma^2$$

Example (Exponential distribution): $X \sim \mathrm{Exp}(\lambda)$

$$f(x) = \lambda e^{-\lambda x}, x > 0$$

Show as exercise (integration by parts) that:

$$E(X) = \int_0^\infty x\lambda e^{-\lambda x} dx = \cdots = \frac{1}{\lambda}$$

Conditional Expectation (Chapter 3)

Suppose we have two random variables X and Y (like in simple linear regression). The conditional expectation of Y given $X = x$ is

$$\mathrm{E}(Y|X = x) = \int y f(y|x) dy$$

where $f(y|x)$ is the conditional density of Y given $X = x$.

Important: Whereas $\mathrm{E}(Y)$ is a number, $\mathrm{E}(Y|X = x)$ is a function of x

If we don't have an observed value for X, then $\mathrm{E}(Y|X)$ is a random variable!

Law of Iterated Expectations:

$$\mathrm{E}[\mathrm{E}(Y|X)] = \mathrm{E}(Y)$$

Convergence

There are several types of convergence of an infinite sequence $\{X_n\} = X_1, X_2, \ldots$ of random variables to another random variable (or constant) X:

- Almost sure (a.s.), also known as strong, which happens when
$$P(\{s: X_n(s) \to X(s)\}) = 1$$
- In probability, also known as weak
- In distribution: $\{X_n\}$ converges in distribution to X if $\mathrm{P}(X_n \leq x) \to \mathrm{P}(X \leq x)$ as $n \to \infty$, for all x
- Notations are $X_n \xrightarrow{\text{a.s.}} X$, $X_n \xrightarrow{P} X$ and $X_n \xrightarrow{d} X$, respectively.
- There is also L_p convergence if $E|X_n - X|^p \to 0$

Quadratic Mean Convergence

- A special case of also L_p convergence, which is defined as $E|X_n - X|^p \to 0$, is the *quadratic mean or* L_2 convergence:

$$X_n \xrightarrow{L_2} X \iff E(X_n - X)^2 \to 0$$

- The L_2 convergence is especially relevant in statistics, because it is related to mean squared error (MSE) of an estimator $\hat{\theta}_n$:

$$\mathrm{MSE}(\hat{\theta}_n) = E\left[\left(\hat{\theta}_n - \theta\right)^2\right]$$

- In these cases, $X_n = \hat{\theta}_n$ is the estimator of a parameter and X is a constant, the parameter θ.

- **Exercise**: Show that $X_n \xrightarrow{L_2} c$ if and only if

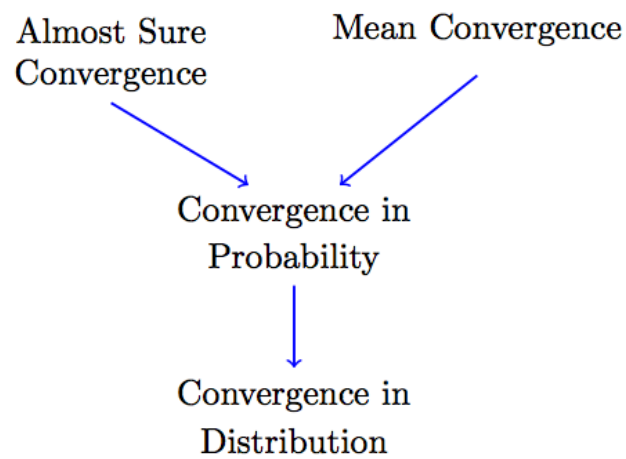
$$\lim_{n\to\infty} E(X_n) \to c \;\; and \;\; \lim_{n\to\infty} Var(X_n) = 0$$

Convergence Properties

- In this course you only need to know the properties of the convergence and related theorems.
- Connection between the different types of convergence:

$$X_n \xrightarrow{\text{a.s.}} X \Rightarrow X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X$$

$$X_n \xrightarrow{\mathrm{L}_2} X \Rightarrow X_n \xrightarrow{\mathrm{L}_1} X \Rightarrow X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X$$



- Special case:

if c is a constant and $X_n \xrightarrow{d} c$, then $X_n \xrightarrow{P} c$

Fundamental Results from Probability Theory

- **Strong Law of Large Numbers (LLN):**

Let $X_1, \ldots, X_n$ be an iid sample from a population with mean $\mu < \infty$. Define the sample mean as $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then:

$$\boxed{\bar{X} \xrightarrow{\text{a.s.}} \mu}$$

- **Central Limit Theorem (CLT):**

Let $X_1, \ldots, X_n$ be an iid sample from a population with standard deviation $\sigma < \infty$. Then:

$$\boxed{\sqrt{n}\frac{\bar{X} - \mu}{\sigma} \xrightarrow{d} N(0,1)}$$

Additional Results

- **Continuous mapping theorem:**

Let $X_n \to X$ and g be an almost everywhere (a.e.) continuous function, then $g(X_n) \to g(X)$.

Note the result is true for any type of convergence.

- **Slutsky's Theorem:**

Let $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, where c is a constant. Then:

a) $X_n + Y_n \xrightarrow{d} X + c$

b) $X_n Y_n \xrightarrow{d} cX$

Example 1

- These results are the backbone of statistics!
- Recall the *t*-test statistic: $T_n = \frac{\bar{X}-\mu}{\frac{S_n}{\sqrt{n}}}$, where $S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$
- Show that $T_n \xrightarrow{d} N(0,1)$
- Proof:

$$T_n = \frac{\bar{X}-\mu}{\frac{S_n}{\sqrt{n}}} = \sqrt{n}\frac{\bar{X}-\mu}{S_n} = \left(\frac{\sigma}{S_n}\right)\left(\sqrt{n}\frac{\bar{X}-\mu}{\sigma}\right)$$

We know by CLT that $\sqrt{n}\frac{\bar{X}-\mu}{\sigma} \xrightarrow{d} N(0,1)$

If we prove that $\frac{\sigma}{S_n} \xrightarrow{P} 1$, that will mean $\frac{\sigma}{S_n} \xrightarrow{d} 1$ and the result follows from Slutsky's Theorem, part b). See Exercise on next page.

Example 2

- Prove that $(\bar{X})^2 \xrightarrow{a.s.} \mu^2$

Proof: Since $g(x) = x^2$ is a continuous function, the result follows from SLLN + continuous mapping theorem.

(On your homework assignments you must justify all steps!)

- **Exercise**: Define the 2nd sample moment as

$$\hat{\mu}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2$$

a) Assuming $E(X^2) < \infty$, prove that

$$\hat{\mu}_2 \to \sigma^2 + \mu^2$$

b) Prove $\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 \to \sigma^2$

The Delta Method

Theorem:

Suppose that

$$\frac{\sqrt{n}(X_n - \mu)}{\sigma} \xrightarrow{d} \mathrm{N}(0, 1)$$

Let g be a differentiable function such that $g'(\mu) \neq 0$.

Then:

$$\frac{\sqrt{n}(g(X_n) - g(\mu))}{|g'(\mu)|\sigma} \xrightarrow{d} \mathrm{N}(0, 1)$$

CLT in Practice

Let $X_1, \ldots, X_n$ be an iid sample from Bernoulli(p) distribution. This means the population mean $\mu = p$ and the population variance is $\sigma^2 = p(1-p)$. The the sample mean is $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i = \hat{p}$, aka the sample proportion.

By the CLT we have that

$$\frac{\hat{p} - \mu}{\sigma/\sqrt{n}} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \xrightarrow{d} N(0,1)$$

But $\sqrt{\hat{p}(1-\hat{p})} \xrightarrow{P} \sqrt{p(1-p)}$ by LLN and the continuous mapping theorem. Therefore

$$\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \times \frac{\sqrt{p(1-p)}}{\sqrt{\hat{p}(1-\hat{p})}} \xrightarrow{d} \mathrm{N}(0,1) \times 1$$

Time to think: Which theorem did we use in the last implication?

CLT in Practice (continued)

The result

$$\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \xrightarrow{d} \mathrm{N}(0,1)$$

can be used to find probabilities like $\hat{p}$ is within 0.01 of p.

Suppose $n = 25$, $\hat{p} = 0.48$. Then $\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.0999$ and

$$P(|\hat{p} - p| \leq 0.01) = P\left(\frac{|\hat{p} - p|}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \leq \frac{0.01}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}\right)$$

$$\approx \mathrm{pnorm}\left(\frac{0.01}{0.0999}\right) - \mathrm{pnorm}\left(-\frac{0.01}{0.0999}\right) = 0.0797$$

Q: Why is it so small?

A: Sample size is not large enough.

These ideas are further explored when we study confidence intervals.